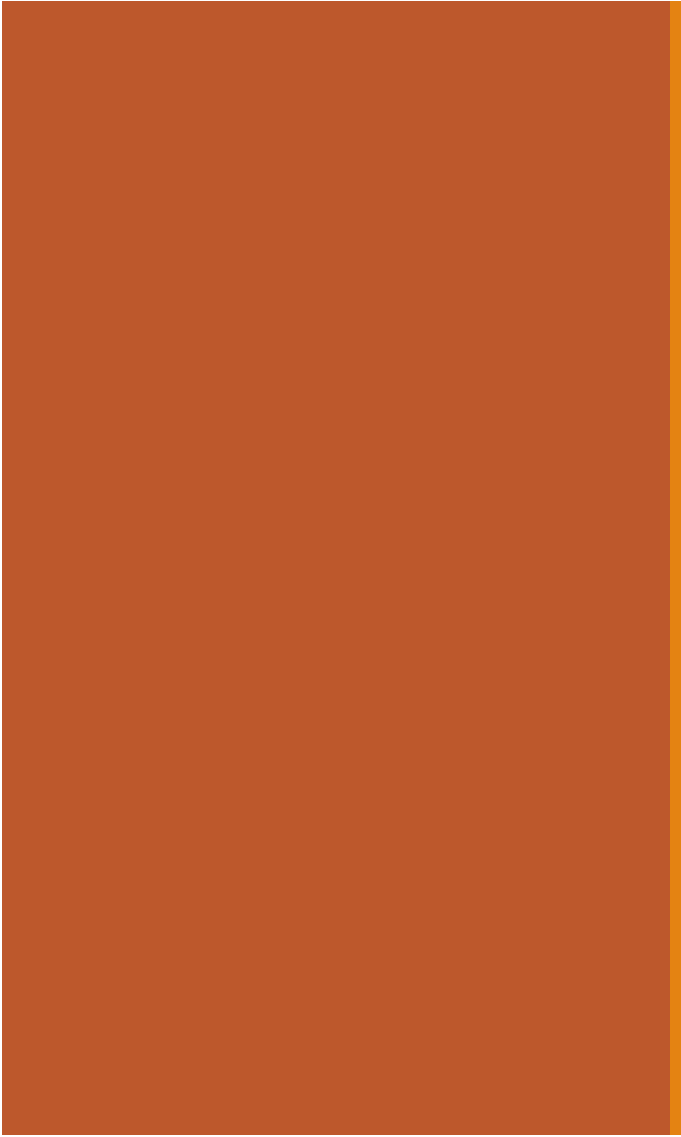


13: Statistics on Multiple Random Variables

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April 29th, 2024

[Lecture Discussion on Ed](#)



Coupon Collecting

Coupon collecting and server requests

The **coupon collector's problem** in probability theory:

- You buy boxes of cereal.
- There are k different types of coupons
- For each box you buy, you "collect" a coupon of type i .

1. How many coupons do you expect after buying n boxes of cereal?



What is the expected number of servers utilized after n requests?

Servers

requests

k servers

request to
server i



- * 52% of Amazon profits
- ** more profitable than Amazon's North America commerce operations

[source](#)

Computer cluster utilization

$$E \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n E[X_i]$$

Consider a computer cluster with k servers. We send n requests.

- Requests independently go to server i with probability p_i
- Let $X = \#$ servers that receive ≥ 1 request.

$$\sum_{i=1}^k p_i = 1$$

What is $E[X]$?



Computer cluster utilization

$$E \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n E[X_i]$$

Consider a computer cluster with k servers. We send n requests.

- Requests independently go to server i with probability p_i
- Let $X = \#$ servers that receive ≥ 1 request.

What is $E[X]$?

1. Define additional random variables.

2. Solve.

Let: A_i = event that server i receives ≥ 1 request

X_i = indicator for A_i

$X_i = \begin{cases} 1 & \text{if } A_i \text{ holds} \\ 0 & \text{if } A_i^c \text{ holds instead} \end{cases}$

$$\begin{aligned} P(A_i) &= 1 - P(\text{no requests to } i) \\ &= 1 - (1 - p_i)^n \end{aligned}$$

$$E[X_i] = P(A_i) = 1 - (1 - p_i)^n$$

$$\begin{aligned} E[X] &= E \left[\sum_{i=1}^k X_i \right] = \sum_{i=1}^k E[X_i] = \sum_{i=1}^k (1 - (1 - p_i)^n) \\ &= \sum_{i=1}^k 1 - \sum_{i=1}^k (1 - p_i)^n = k - \sum_{i=1}^k (1 - p_i)^n \end{aligned}$$

does this result make sense when $n=0$? when $n=1$?

Note: A_i are dependent!

Coupon collecting problems: Hash tables

The **coupon collector's problem** in probability theory:

- You buy boxes of cereal.
- There are k different types of coupons
- For each box you buy, you "collect" a coupon of type i .

1. How many coupons do you expect after buying n boxes of cereal?



What is the expected number of utilized servers after n requests?

2. How many boxes do you expect to buy until you have one of each coupon?



What is the expected number of strings to hash until each bucket has ≥ 1 string?

<u>Servers</u>	<u>Hash Tables</u>
requests	strings
k servers	k buckets
request to server i	hashed to bucket i

Hash Tables

$$E \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n E[X_i]$$

Consider a hash table with k buckets.

- Strings are equally likely to get hashed into any bucket (independently).
- Let $Y = \#$ strings to hash until each bucket ≥ 1 string.

What is $E[Y]$?

1. Define additional random variables.

How should we define Y_i such that $Y = \sum_i Y_i$?

2. Solve.

Hash Tables

$$E \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n E[X_i]$$

assume ideal hash function, so that $p_i = \frac{1}{k}$

Consider a hash table with k buckets.

- Strings are equally likely to get hashed into any bucket (independently).
- Let $Y = \#$ strings to hash until each bucket ≥ 1 string.

What is $E[Y]$?

$Y_0 = \#$ hashes needed until first bucket gets a string
 $Y_1 = \#$ hashes needed until second bucket gets a string
 $Y_2 = \#$ hashes needed until third bucket gets a string

1. Define additional random variables.

Let: $Y_i = \#$ of trials needed to get success after i -th success

- Success: hash string to previously empty bucket
- If i non-empty buckets: $P(\text{success}) = \frac{k-i}{k}$ } numerator is number of empty buckets.

2. Solve.

$$P(Y_i = n) = \left(\frac{i}{k} \right)^{n-1} \left(\frac{k-i}{k} \right)$$

$$\text{Equivalently, } Y_i \sim \text{Geo} \left(p = \frac{k-i}{k} \right) \quad E[Y_i] = \frac{1}{p} = \frac{k}{k-i}$$

Hash Tables

$$E \left[\sum_{i=1}^n X_i \right] = \sum_{i=1}^n E[X_i]$$

Consider a hash table with k buckets.

- Strings are equally likely to get hashed into any bucket (independently).
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What is $E[Y]$?

1. Define additional random variables. Let: $Y_i = \#$ of trials to needed get success after i -th success

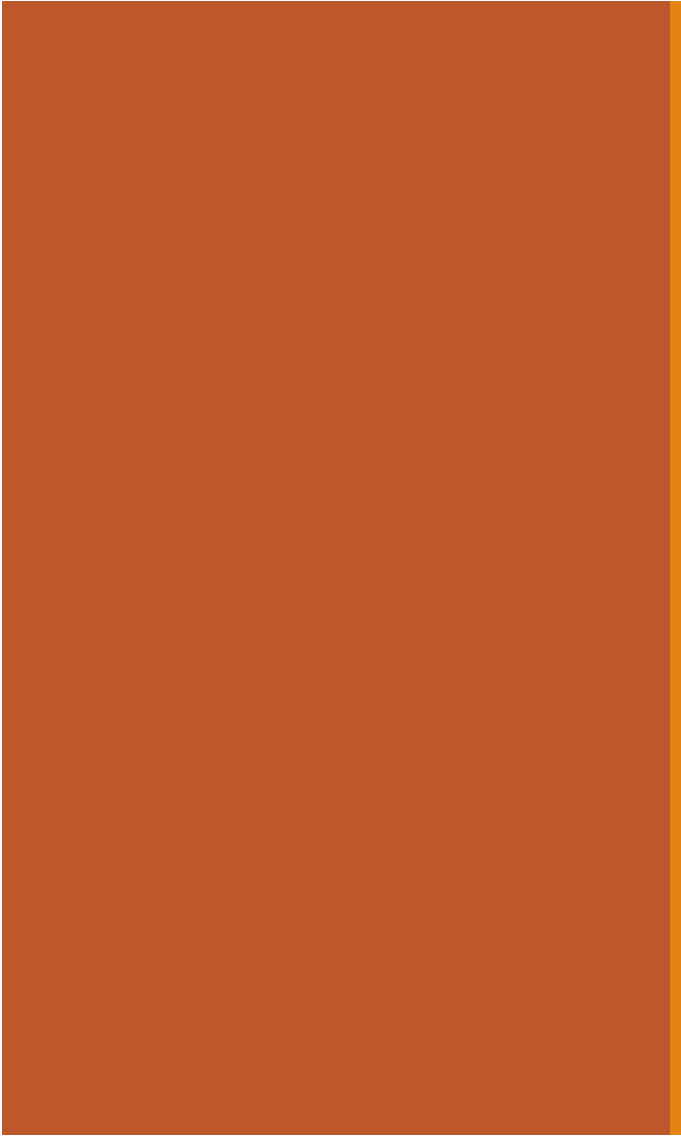
$$Y_i \sim \text{Geo} \left(p = \frac{k-i}{k} \right), \quad E[Y_i] = \frac{1}{p} = \frac{k}{k-i}$$

$$\sum_{m=1}^k \frac{1}{m} \approx \int_1^k \frac{1}{m} dm = \ln k$$

2. Solve. $Y = Y_0 + Y_1 + \dots + Y_{k-1}$

$$E[Y] = E[Y_0] + E[Y_1] + \dots + E[Y_{k-1}]$$

$$= \frac{k}{k} + \frac{k}{k-1} + \frac{k}{k-2} + \dots + \frac{k}{1} = k \left[\frac{1}{k} + \frac{1}{k-1} + \dots + 1 \right] = O(k \log k)$$



Covariance

Statistics of sums of RVs

For any random variables X and Y ,

$$E[X + Y] = E[X] + E[Y]$$

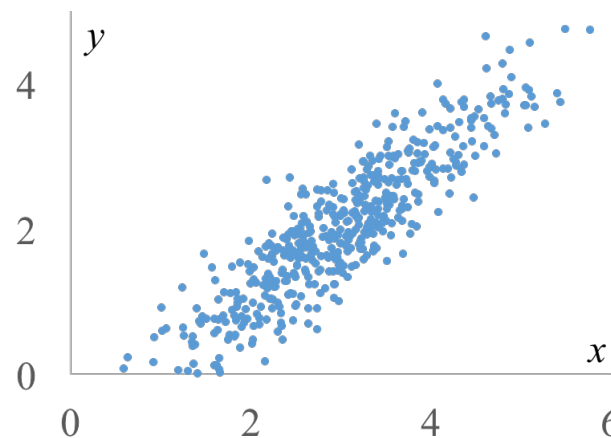
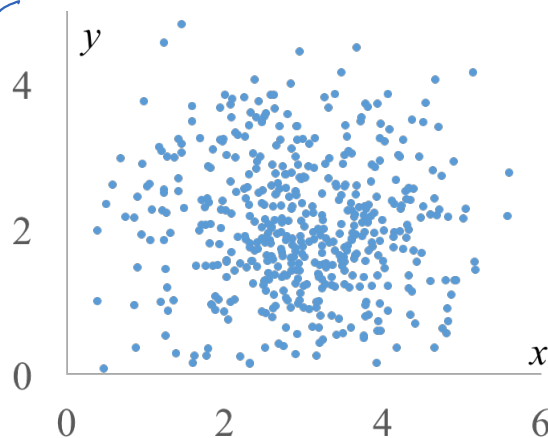
$$\text{Var}(X + Y) = \quad ?$$

But first, a new statistic!

Spot the difference

Compare/contrast the following two distributions:

on the left, you can't easily predict how Y will change as x increases.



Assume all points are equally likely.

$$P(X = x, Y = y) = \frac{1}{N}$$

on the right, you can predict that, as x increases, y increases as well.

these statistics don't really communicate how X and Y are coupled.

Both distributions have the same $E[X]$, $E[Y]$, $\text{Var}(X)$, and $\text{Var}(Y)$

Difference: how the two variables vary with **each other**.

Covariance

The **covariance** of two variables X and Y is:

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

Proof of second part (rewriting $E[X]$, $E[Y]$ as μ_X , μ_Y to emphasize that they're each constants):

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] = E[(X - \mu_X)(Y - \mu_Y)] \\ &= E[XY - \mu_Y X - \mu_X Y + \mu_X \mu_Y] \\ &= E[XY] - E[\mu_Y X] - E[\mu_X Y] + E[\mu_X \mu_Y] && \text{(linearity of expectation)} \\ &= E[XY] - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y && (\mu_X, \mu_Y \text{ are constants}) \\ &= E[XY] - \mu_X \mu_Y = E[XY] - E[X]E[Y]\end{aligned}$$

Covariance

The **covariance** of two variables X and Y is:

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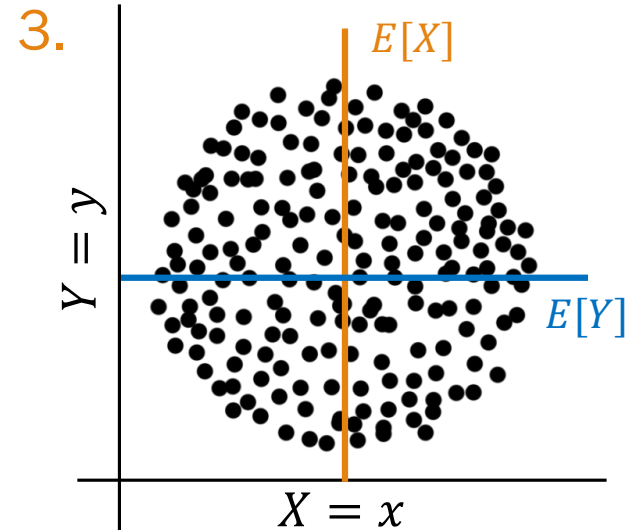
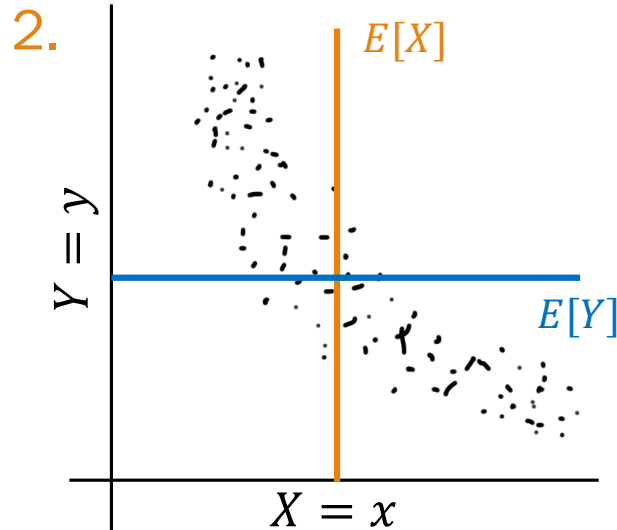
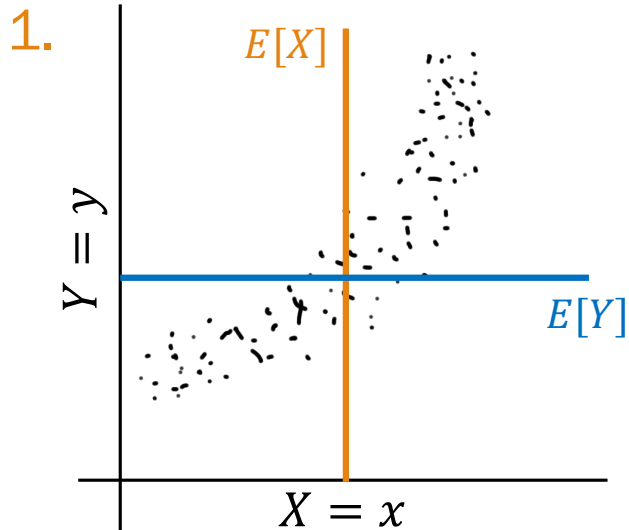
Covariance measures how one random variable varies with a second.

- Outside temperature and utility bills have a **negative** covariance.
- Handedness and musical ability have near **zero** covariance.
- Product demand and price have a **positive** covariance.

Feel the covariance

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

Is the covariance positive, negative, or zero?

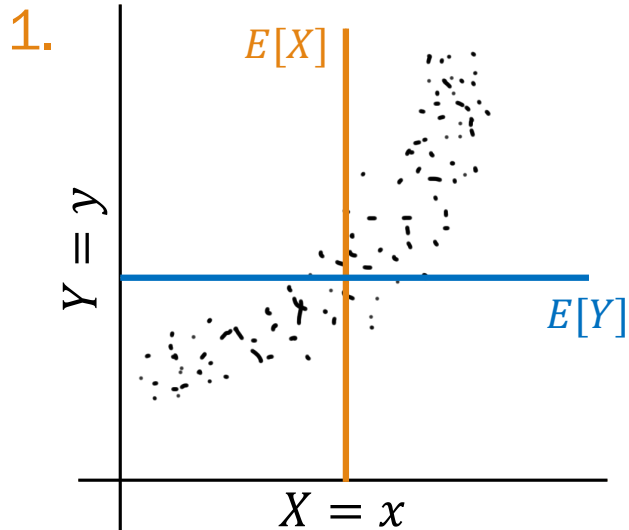


Feel the covariance

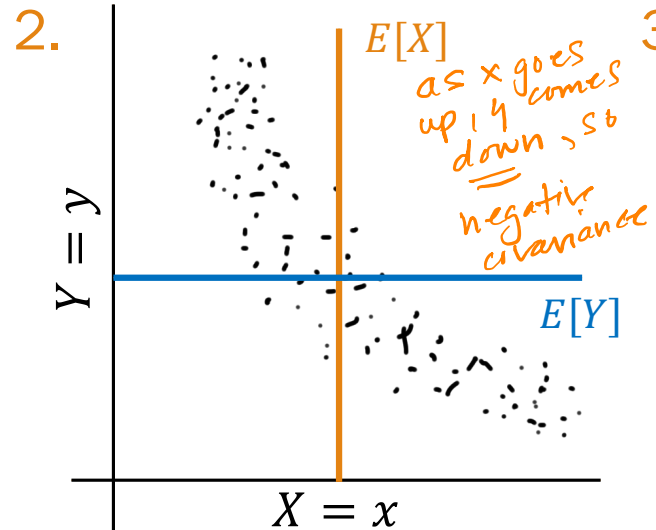
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Is the covariance positive, negative, or zero?

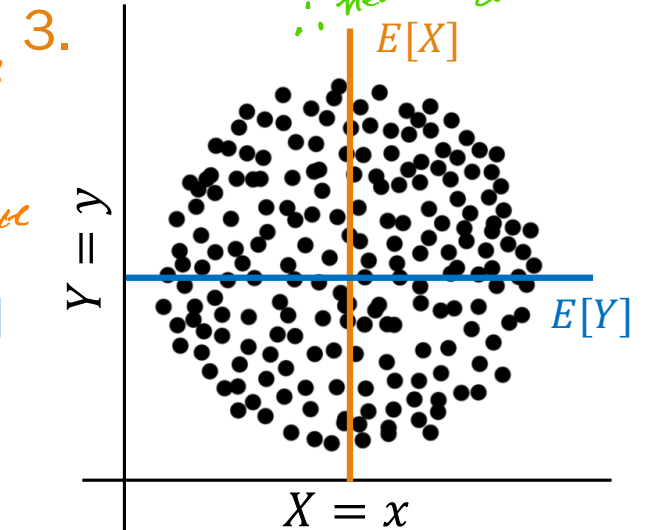
as x increases, so does y: positive covariance



positive



negative



zero

Covarying humans

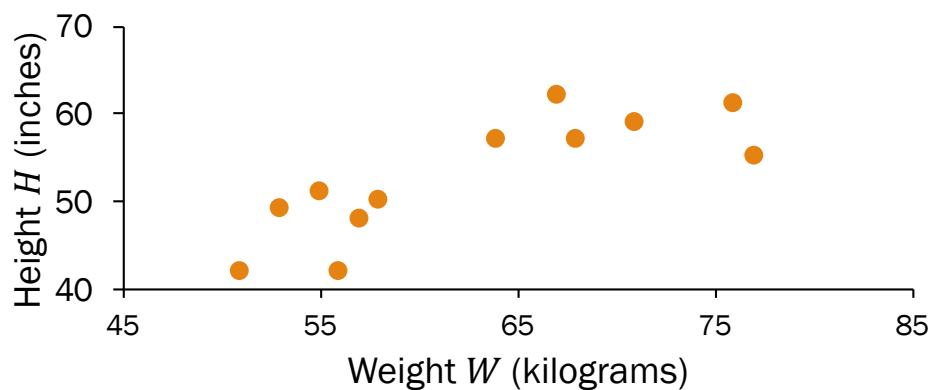
$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

Weight (kg)	Height (in)	W · H
64	57	3648
71	59	4189
53	49	2597
67	62	4154
55	51	2805
58	50	2900
77	55	4235
57	48	2736
56	42	2352
51	42	2142
76	61	4636
68	57	3876

$$\begin{aligned}E[W] &= 62.75 \\ E[H] &= 52.75 \\ E[WH] &= 3355.83\end{aligned}$$

What is the covariance of weight W and height H ?

$$\begin{aligned}\text{Cov}(W, H) &= E[WH] - E[W]E[H] \\ &= 3355.83 - (62.75)(52.75) \\ (\text{positive}) &= 45.77\end{aligned}$$



Covariance > 0: one variable ↑, other variable ↑

Properties of Covariance

The covariance of two variables X and Y is:

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

Properties:

1. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$
2. $\text{Var}(X) = E[X^2] - (E[X])^2 = E[XX] - E[X]E[X] = \text{Cov}(X, X)$
3. Covariance of sums = sum of all pairwise covariances (proof left to you)
 $\text{Cov}(X_1 + X_2, Y_1 + Y_2) = \text{Cov}(X_1, Y_1) + \text{Cov}(X_2, Y_1) + \text{Cov}(X_1, Y_2) + \text{Cov}(X_2, Y_2)$
4. Covariance under linear transformation: $\text{Cov}(aX + b, Y) = a\text{Cov}(X, Y)$
recall that $\text{Var}(aX+b) = a^2\text{Var}(X)$! this seems consistent with that,

Zero covariance does not imply independence

Let X take on values $\{-1, 0, 1\}$
with equal probability $1/3$.

Define $Y = \begin{cases} 1 & \text{if } X = 0 \\ 0 & \text{otherwise} \end{cases}$

*Y is intentionally
defined so that
X is 0 iff
Y is nonzero*

What is the joint PMF of X and Y ?

Zero covariance does not imply independence

Let X take on values $\{-1, 0, 1\}$ with equal probability $1/3$.

Define $Y = \begin{cases} 1 & \text{if } X = 0 \\ 0 & \text{otherwise} \end{cases}$

		X			
		-1	0	1	
Y	0	1/3	0	1/3	2/3
	1	0	1/3	0	1/3
		1/3	1/3	1/3	

Marginal PMF of Y , $p_Y(y)$

Marginal PMF of X , $p_X(x)$

1. $E[X] =$

$E[Y] =$

2. $E[XY] =$

3. $\text{Cov}(X, Y) =$

4. Are X and Y independent?



Zero covariance does not imply independence

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Marginal PMF of Y , $p_Y(y)$

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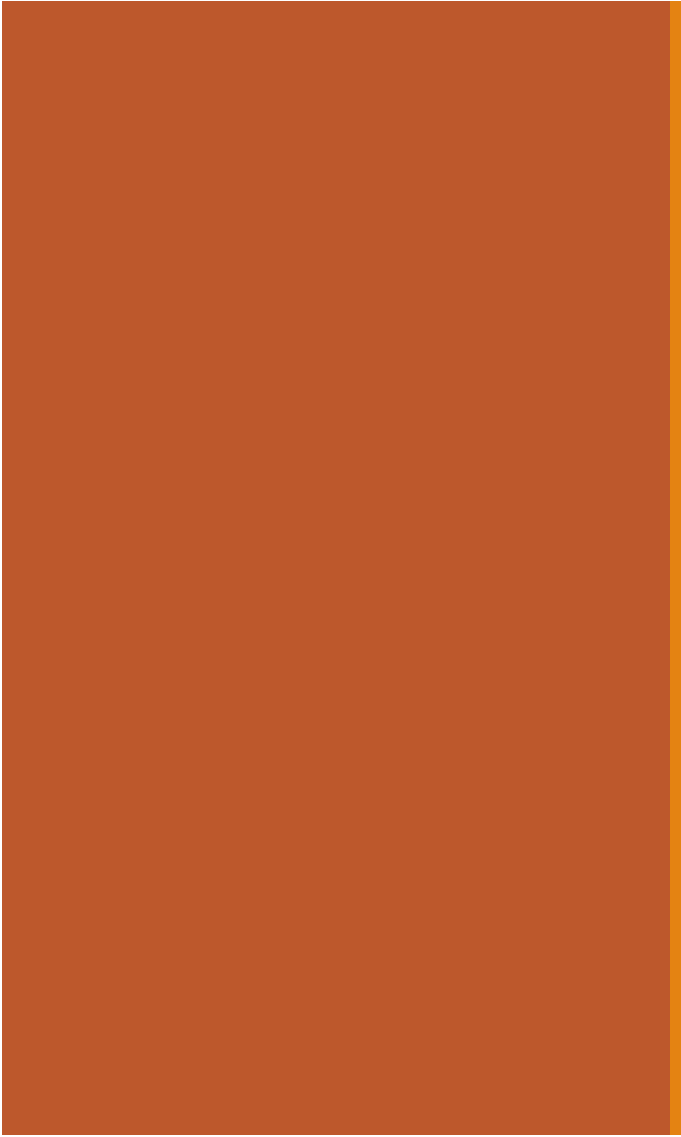
$$1. \quad E[X] = -1\left(\frac{1}{3}\right) + 0\left(\frac{1}{3}\right) + 1\left(\frac{1}{3}\right) = 0 \quad E[Y] = 0\left(\frac{2}{3}\right) + 1\left(\frac{1}{3}\right) = 1/3$$

$$2. \quad E[XY] = (-1 \cdot 0)\left(\frac{1}{3}\right) + (0 \cdot 1)\left(\frac{1}{3}\right) + (1 \cdot 0)\left(\frac{1}{3}\right) = 0$$

$$3. \quad \text{Cov}(X, Y) = E[XY] - E[X]E[Y] = 0 - 0(1/3) = 0 \quad \text{! does not imply independence!}$$

4. Are X and Y independent? **✗**

$$P(Y = 0 | X = 1) = 1 \neq P(Y = 0) = 2/3$$



Variance of sums of RVs

Statistics of sums of RVs

For any random variables X and Y ,

$$E[X + Y] = E[X] + E[Y]$$

$$\text{Var}(X + Y) = \text{Var}(X) + 2 \cdot \text{Cov}(X, Y) + \text{Var}(Y)$$

Variance of general sum of RVs

For any random variables X and Y ,

$$\text{Var}(X + Y) = \text{Var}(X) + 2 \cdot \text{Cov}(X, Y) + \text{Var}(Y)$$

Proof:

$$\text{Var}(X + Y) = \text{Cov}(X + Y, X + Y)$$

$$\text{Var}(X) = \text{Cov}(X, X)$$

$$= \text{Cov}(X, X) + \text{Cov}(X, Y) + \text{Cov}(Y, X) + \text{Cov}(Y, Y)$$

covariance of
all pairs

$$= \text{Var}(X) + \underbrace{2 \cdot \text{Cov}(X, Y)}_{\substack{\text{same} \\ \text{symmetry}}} + \text{Var}(Y)$$

Symmetry of covariance +
 $\text{Cov}(X, X) = \text{Var}(X)$

More generally:

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n \text{Cov}(X_i, X_j) \quad (\text{proof in extra slides})$$

Statistics of sums of RVs

For any random variables X and Y ,

$$E[X + Y] = E[X] + E[Y]$$

$$\text{Var}(X + Y) = \text{Var}(X) + 2 \cdot \text{Cov}(X, Y) + \text{Var}(Y)$$

For **independent** X and Y ,

$$E[XY] = E[X]E[Y]$$

(Lemma: proof in extra slides)

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

Variance of sum of independent RVs

For **independent** X and Y ,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

Proof:

1. $\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$

$$= E[X]E[Y] - E[X]E[Y]$$

$$= 0$$

def. of covariance

X and Y are **independent**

this is 0 when X and Y are independent.

2. $\text{Var}(X + Y) = \text{Var}(X) + 2 \cdot \text{Cov}(X, Y) + \text{Var}(Y)$

$$= \text{Var}(X) + \text{Var}(Y)$$

Proving Variance of the Binomial

$$X \sim \text{Bin}(n, p) \quad \text{Var}(X) = np(1 - p)$$

Let $X = \sum_{i=1}^n X_i$

Let $X_i = i\text{th trial is heads}$
 $X_i \sim \text{Ber}(p)$
 $\text{Var}(X_i) = p(1 - p)$

X_i are **independent**
(by definition)

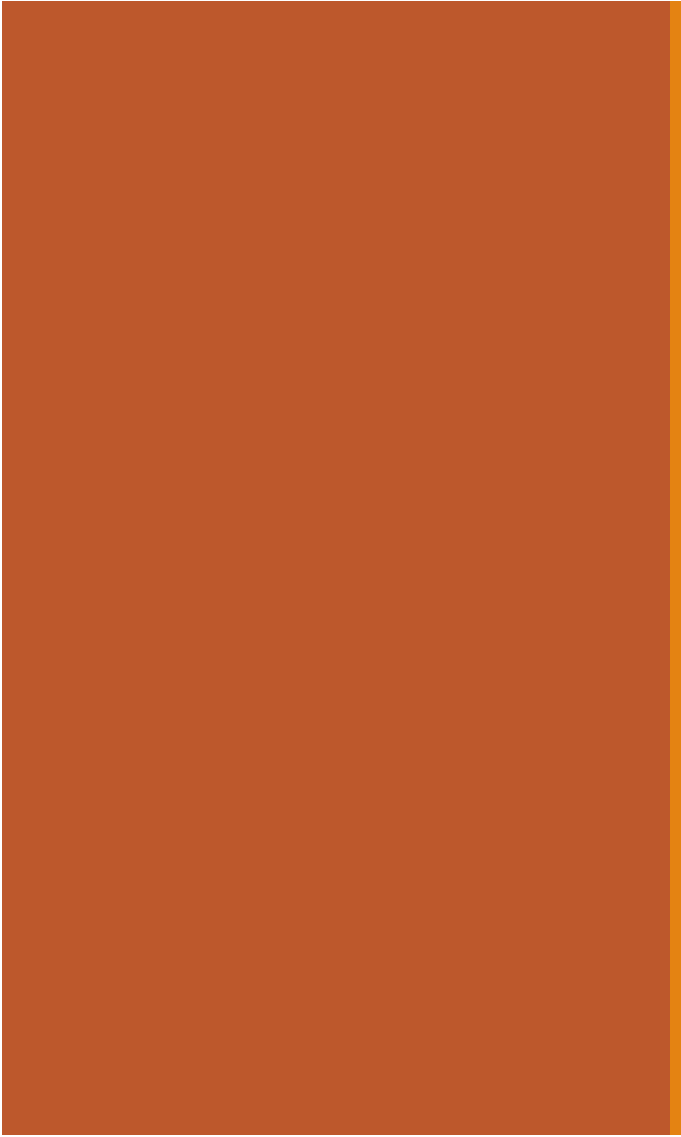
$$\begin{aligned} \text{Var}(X) &= \text{Var}\left(\sum_{i=1}^n X_i\right) \\ &= \sum_{i=1}^n \text{Var}(X_i) \\ &= \sum_{i=1}^n p(1 - p) \\ &= np(1 - p) \end{aligned}$$

X_i are **independent**,
therefore variance of sum
= sum of variance

Variance of Bernoulli

yay!





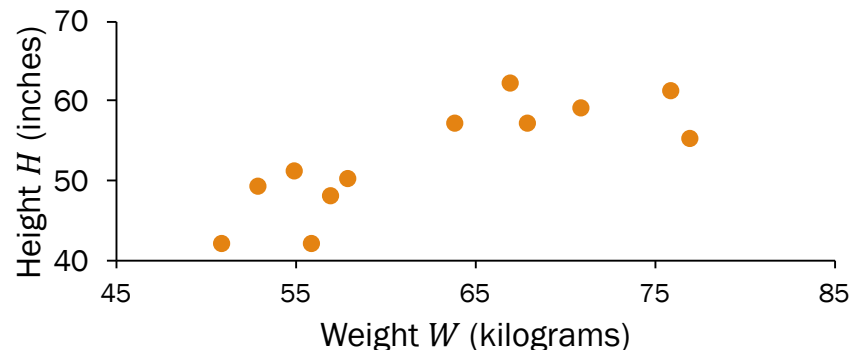
Correlation

Covarying humans

$$\begin{aligned}\text{Cov}(X, Y) &= E[(X - E[X])(Y - E[Y])] \\ &= E[XY] - E[X]E[Y]\end{aligned}$$

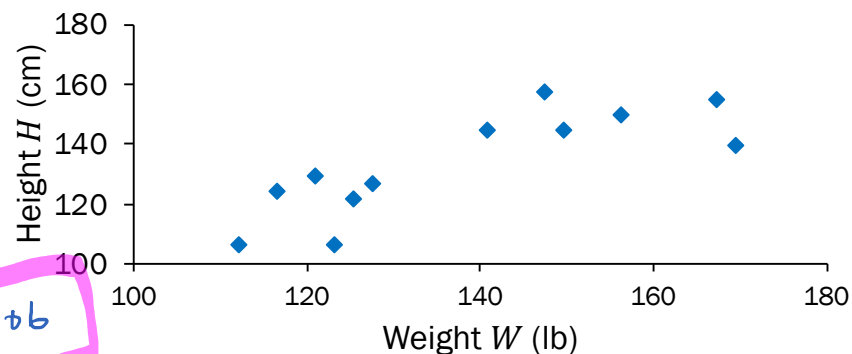
What is the covariance of weight W and height H ?

$$\begin{aligned}\text{Cov}(W, H) &= E[WH] - E[W]E[H] \\ &= 3355.83 - (62.75)(52.75) \\ &= 45.77 \text{ (positive)}\end{aligned}$$



What about weight (lb) and height (cm)?

$$\begin{aligned}\text{Cov}(2.20W, 2.54H) &= E[2.20W \cdot 2.54H] - E[2.20W]E[2.54H] \\ &= 18752.38 - (138.05)(133.99) \\ &= 255.06 \text{ (positive)}\end{aligned}$$



Covariance depends on units!

$$2.20 \cdot 2.54 \cdot 45.77 \approx 255.06$$

45.77

255.06

Sign of covariance (+/-) more meaningful than magnitude

Correlation

The **correlation** of two variables X and Y is:

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$

$$\begin{aligned}\sigma_X^2 &= \text{Var}(X), \\ \sigma_Y^2 &= \text{Var}(Y)\end{aligned}$$

- Note: $-1 \leq \rho(X, Y) \leq 1$
- Correlation measures the **linear relationship** between X and Y :

$$\rho(X, Y) = 1 \quad \Rightarrow Y = aX + b, \text{ where } a = \sigma_Y / \sigma_X$$

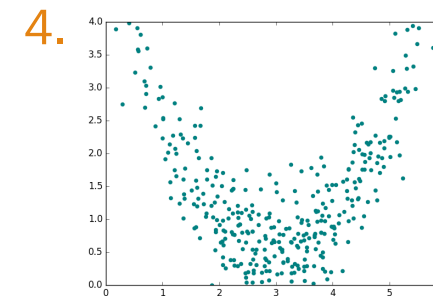
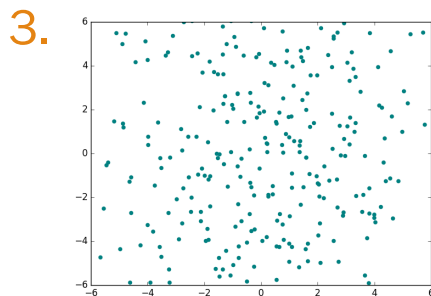
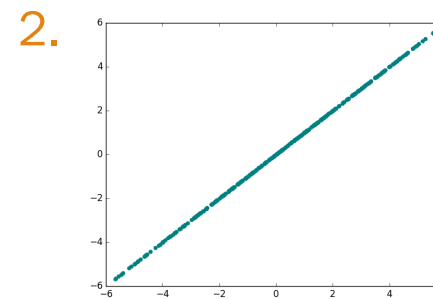
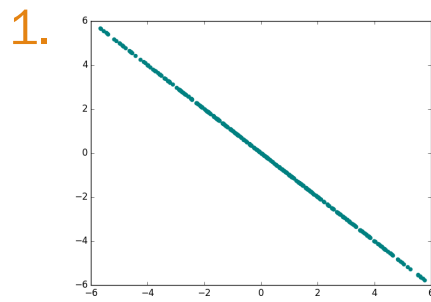
$$\rho(X, Y) = -1 \quad \Rightarrow Y = aX + b, \text{ where } a = -\sigma_Y / \sigma_X$$

$$\rho(X, Y) = 0 \quad \Rightarrow \text{uncorrelated (absence of linear relationship)}$$

Correlation reps

What is the correlation coefficient $\rho(X, Y)$?

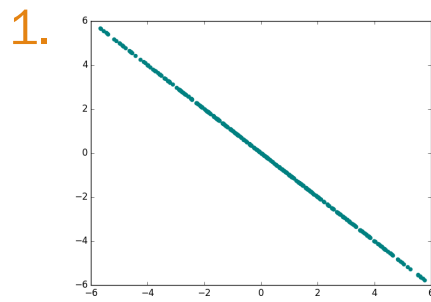
- A. $\rho(X, Y) = 1$
- B. $\rho(X, Y) = -1$
- C. $\rho(X, Y) = 0$
- D. Other



Correlation reps

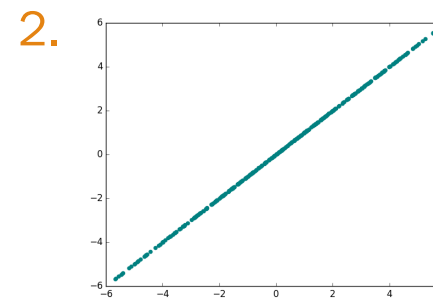
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- D. Other



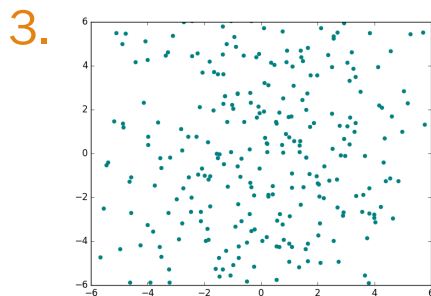
B. $\rho(X, Y) = -1$

$$Y = -aX + b$$
$$a > 0$$



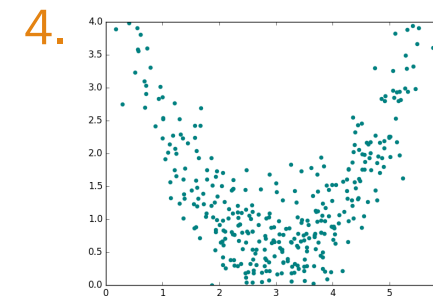
A. $\rho(X, Y) = 1$

$$Y = aX + b$$
$$a > 0$$



C. $\rho(X, Y) = 0$

“uncorrelated”

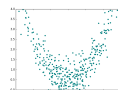


C. $\rho(X, Y) = 0$

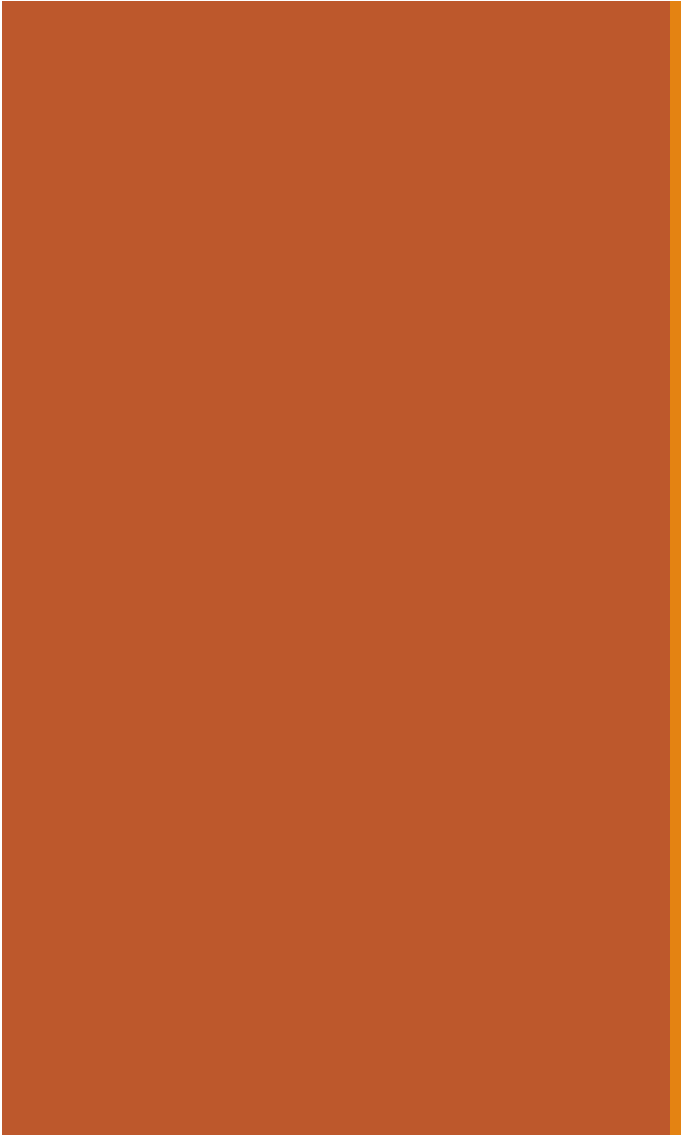
$$Y = X^2$$

X and Y can be nonlinearly related even if $\rho(X, Y) = 0$.

Throwback to CS103: Conditional statements

Statement $P \rightarrow Q$:	Independence $\rightarrow$ No correlation	✓
Contrapositive $\neg Q \rightarrow \neg P$:	Correlation $\rightarrow$ Dependence	✓ (logically equivalent)
Inverse $\neg P \rightarrow \neg Q$:	Dependence $\rightarrow$ Correlation	✗ (not always) $Y = X^2$ $\rho(X, Y) = 0$ 
Converse $Q \rightarrow P$:	No correlation $\rightarrow$ Independence	✗ (not always)

“Correlation does not imply causation”

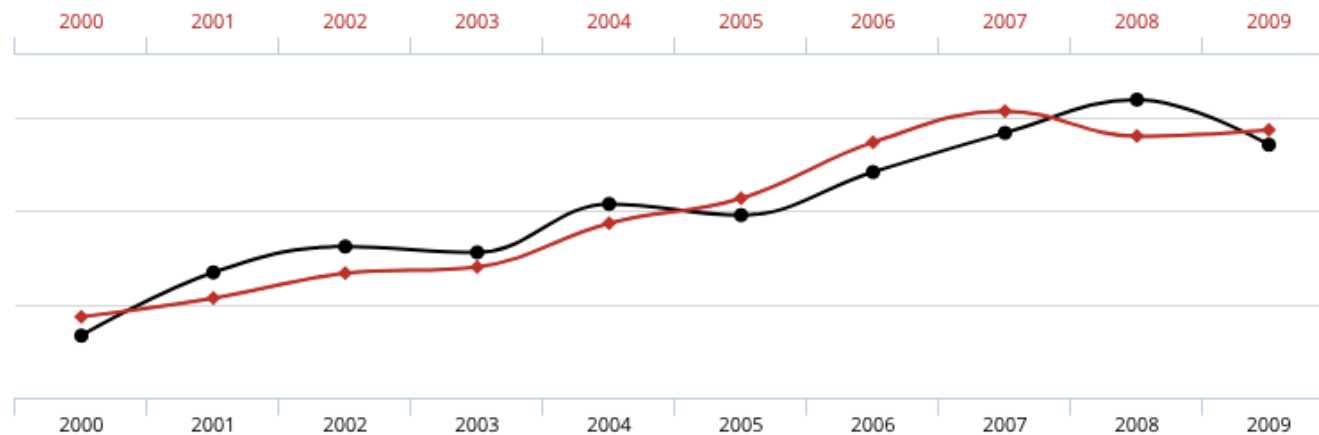


Spurious Correlation

Spurious Correlations

$\rho(X, Y)$ is used a lot to statistically quantify the relationship b/t X and Y.

Correlation:
0.947091



Spurious Correlations

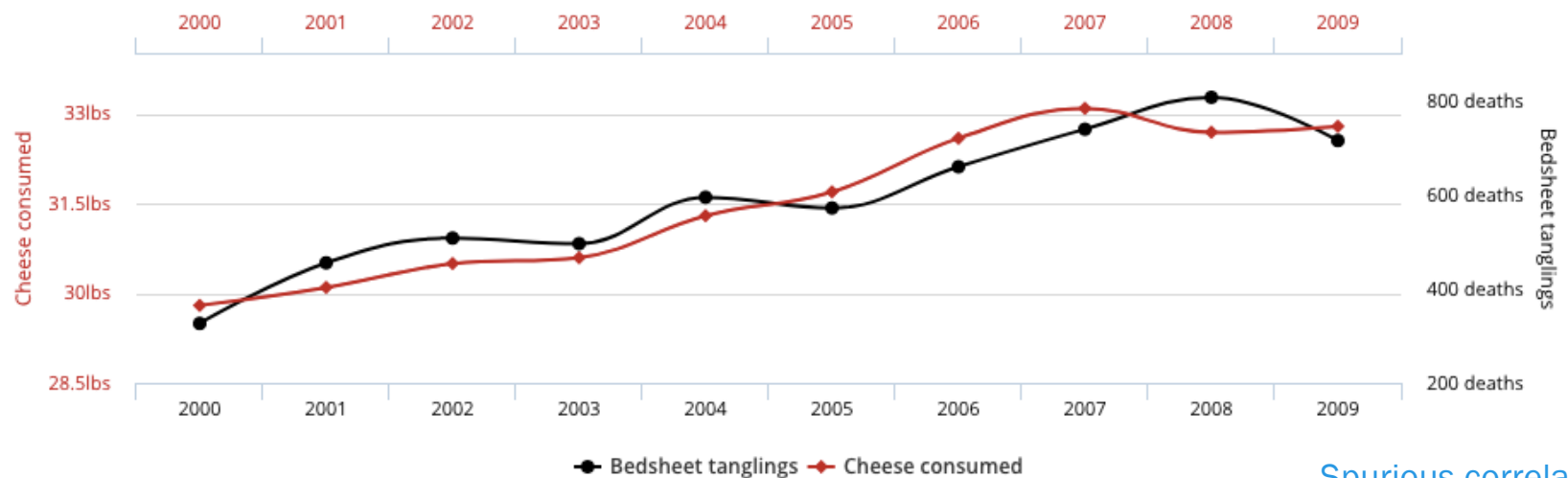
$\rho(X, Y)$ is used a lot to statistically quantify the relationship b/t X and Y.

Correlation:
0.947091

Per capita cheese consumption

correlates with

Number of people who died by becoming tangled in their bedsheets

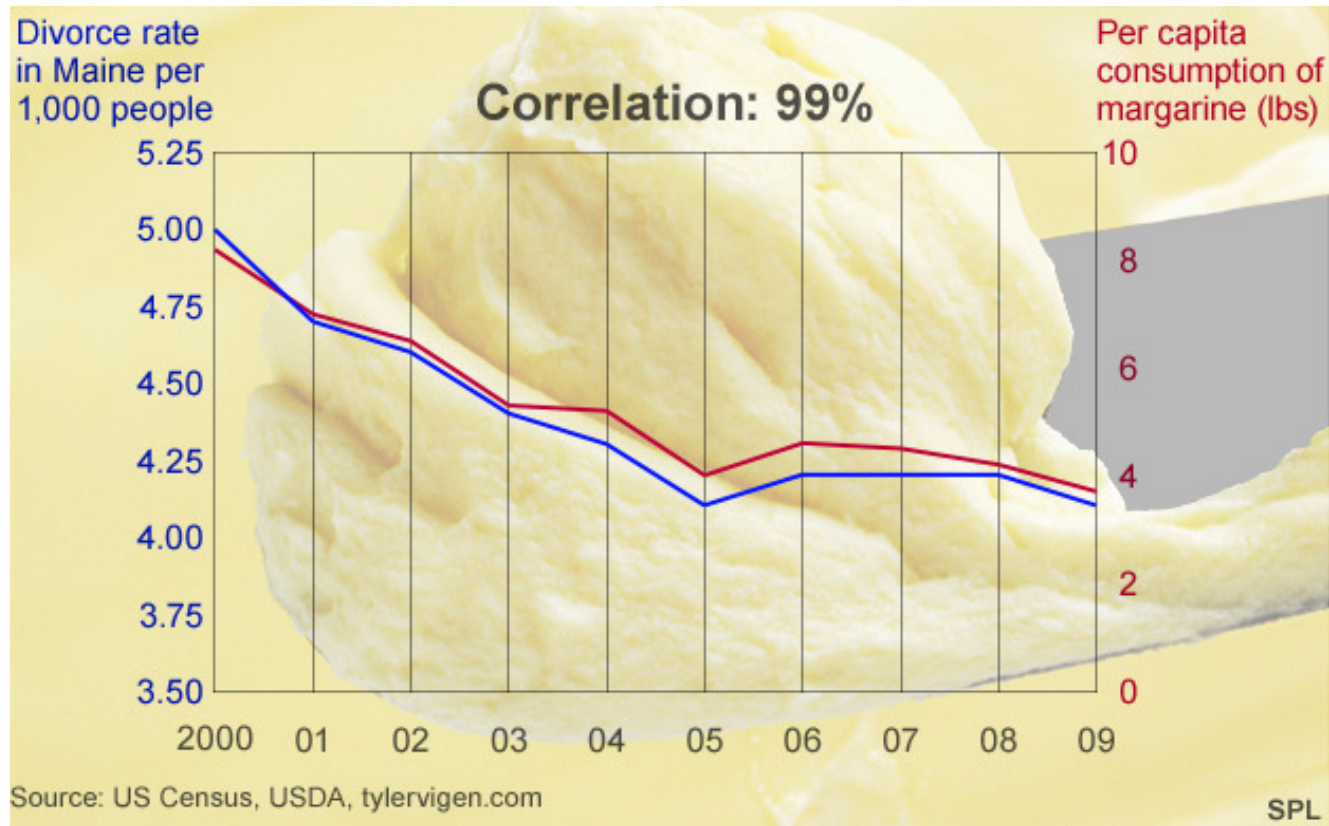


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Divorce vs. Margarine



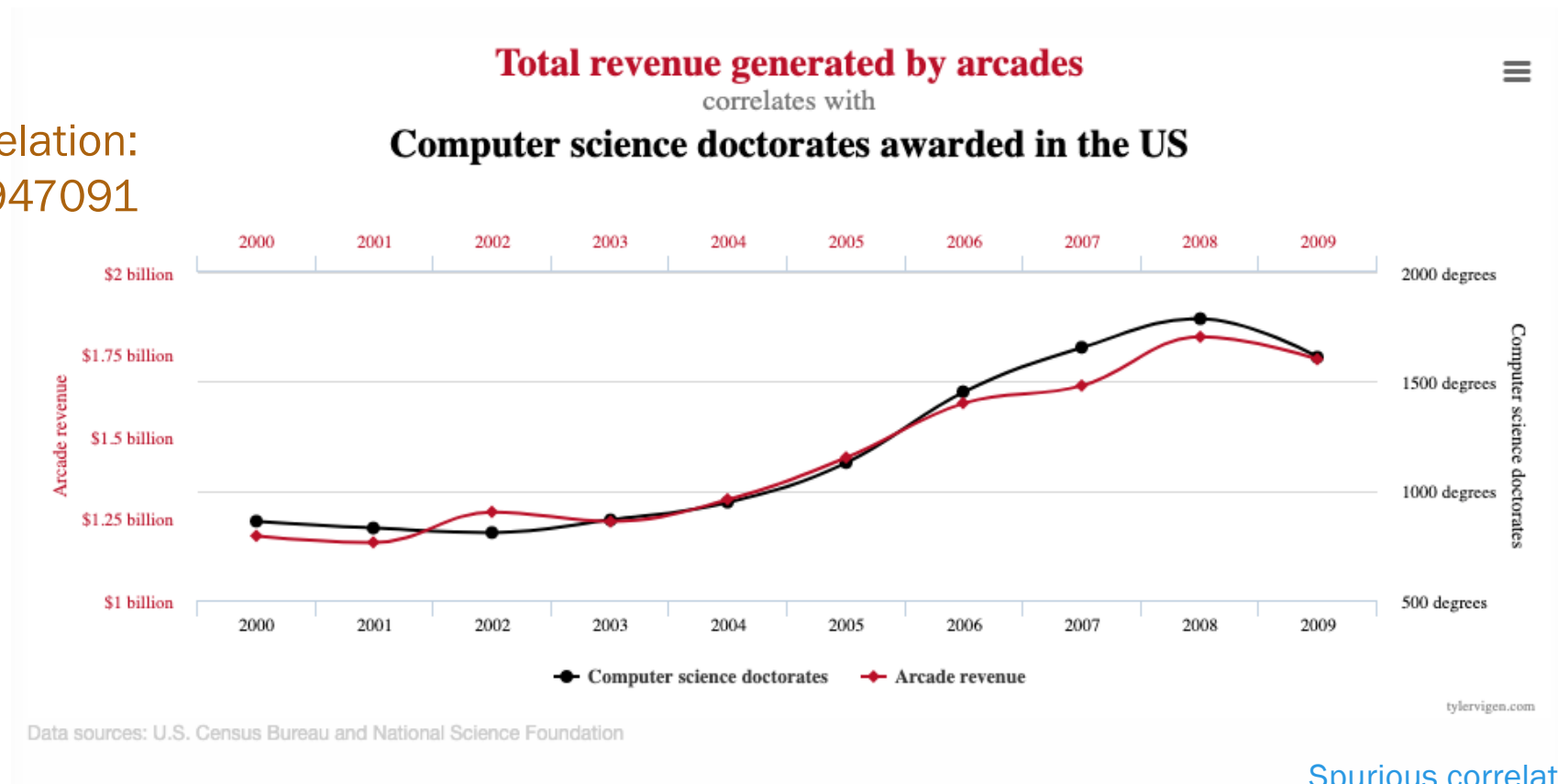
<http://www.bbc.com/news/magazine-27537142>

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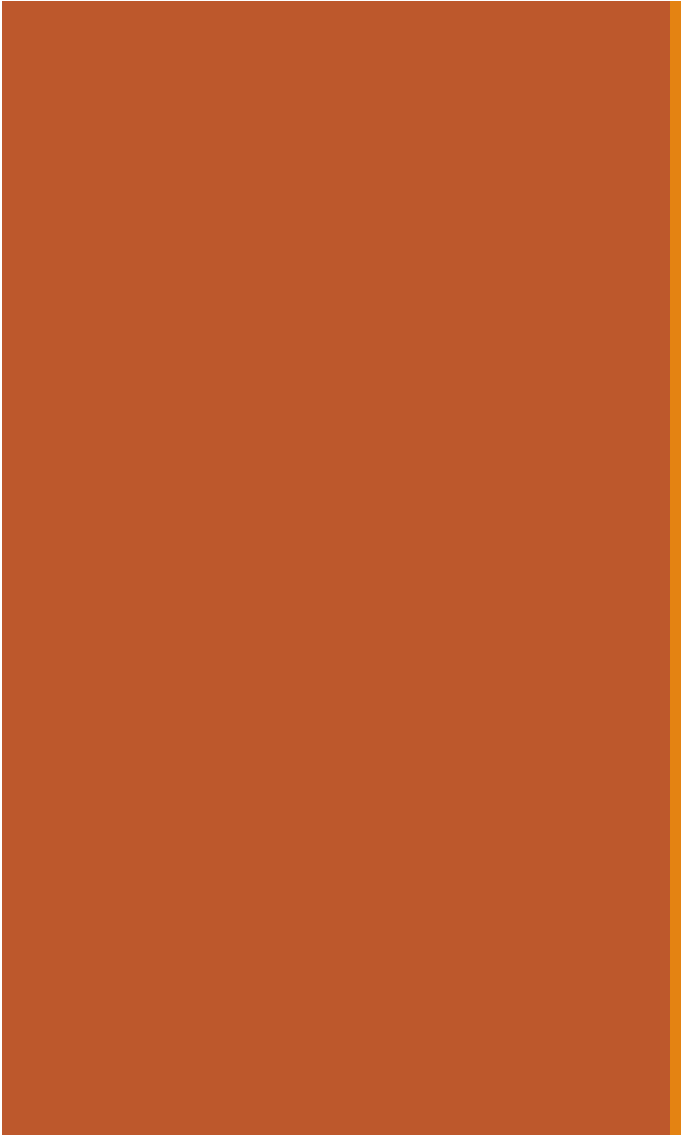
Arcade revenue vs. CS PhDs

Correlation:
0.947091



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A large, solid orange rectangle occupies the left side of the slide, extending from the top to the bottom and from the left edge to about one-third of the way across the slide.

Extras

Expectation of product of independent RVs

If X and Y are
independent, then

$$\begin{aligned} E[XY] &= E[X]E[Y] \\ E[g(X)h(Y)] &= E[g(X)]E[h(Y)] \end{aligned}$$

Proof: $E[g(X)h(Y)] = \sum_y \sum_x g(x)h(y)p_{X,Y}(x,y)$ (for continuous proof, replace summations with integrals)

$= \sum_y \sum_x g(x)h(y)p_X(x)p_Y(y)$ X and Y are independent

$= \sum_y \left(h(y)p_Y(y) \sum_x g(x)p_X(x) \right)$ Terms dependent on y are constant in integral of x

$= \left(\sum_x g(x)p_X(x) \right) \left(\sum_y h(y)p_Y(y) \right)$ Summations separate

$= E[g(X)]E[h(Y)]$

Variance of Sums of Variables

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n \text{Cov}(X_i, X_j)$$

Proof:

$$\begin{aligned} \text{Var}\left(\sum_{i=1}^n X_i\right) &\stackrel{\text{Var}(X) = \text{Cov}(X, X)}{=} \text{Cov}\left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i\right) \stackrel{\text{covariance of all pairs}}{=} \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j) \\ &= \sum_{i=1}^n \text{Var}(X_i) + \sum_{i=1}^n \sum_{j=1, j \neq i}^n \text{Cov}(X_i, X_j) \\ &= \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=i+1}^n \text{Cov}(X_i, X_j) \end{aligned}$$

Symmetry of covariance
 $\text{Cov}(X, X) = \text{Var}(X)$

Adjust summation bounds